

SQL INTERVIEW PREP

20 TRICKY SQL JOIN PATTERNS

Common Traps & Advanced Logic

1. Fill Missing Dates

2. Team Pairings

3. Bought A AND B

4. Bought A NOT B

5. Same Birth Month

6. Update via Join

7. Full Join Union

8. Symmetric Diff

9. Running Total

10. "Top 1" Join

11. Null Key Join

12. LIKE Join

13. Delete Duplicates

14. Many-to-Many

15. Recursive CTE

16. Lateral / Apply

17. Conditional Join

18. Having vs Where

19. Join vs Exists

20. Fan-Out Trap

Q1

FILL MISSING DATES

Report 0 for days with no sales



Standard SQL

-- Generate all Date x Product combos

```
SELECT d.date, p.product,  
        COALESCE(s.sales, 0)  
FROM calendar d  
CROSS JOIN products p  
LEFT JOIN sales s  
        ON d.date = s.date AND p.id = s.pid;
```



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Q2

TEAM PAIRINGS

Unique matches (A vs B, not B vs A)



Standard SQL

-- Use inequality to avoid mirrors

```
SELECT a.team_name, b.team_name
FROM teams a
JOIN teams b
  ON a.id < b.id;
```



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Q3

BOUGHT A AND B

Customers who bought both items



Standard SQL

-- Self Join on Customer ID

```
SELECT t1.cust_id
FROM orders t1
JOIN orders t2
  ON t1.cust_id = t2.cust_id
WHERE t1.item = 'Milk'
  AND t2.item = 'Bread';
```



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Q4

BOUGHT A NOT B

Bought Phone but NO Cover



Standard SQL

-- Left Join + Is Null

```
SELECT t1.cust_id
FROM orders t1
LEFT JOIN orders t2
  ON t1.cust_id = t2.cust_id
  AND t2.item = 'Cover'
WHERE t1.item = 'Phone'
  AND t2.cust_id IS NULL;
```



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Q5

SAME BIRTH MONTH

Find emps who share birthdays

Standard SQL

-- Join on Month part of date

```
SELECT a.name, b.name
FROM employees a
JOIN employees b
  ON MONTH(a.dob) = MONTH(b.dob)
 AND a.id <> b.id;
```



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Q6

UPDATE VIA JOIN

Deduct stock based on Sales



SQL Server / MySQL

-- Update Target joined with Source

```
UPDATE i
SET i.qty = i.qty - s.sold_qty
FROM inventory i
JOIN daily_sales s
  ON i.product_id = s.product_id;
```



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Q7

FULL JOIN SIM

For DBs without FULL OUTER (MySQL)

MySQL

-- Left Join UNION Right Join

SELECT * FROM a LEFT JOIN b ON a.id=b.id

UNION

SELECT * FROM a RIGHT JOIN b ON a.id=b.id;



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Q8

SYMMETRIC DIFF

Items in A or B, but NOT both



Standard SQL

-- Full Join where one side is NULL

```
SELECT *  
FROM table_a a  
FULL OUTER JOIN table_b b  
  ON a.id = b.id  
WHERE a.id IS NULL  
      OR b.id IS NULL;
```



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Q9

RUNNING TOTAL

The Self-Join Method



Standard SQL

```
-- Inequality Join (Expensive!)  
SELECT a.date, SUM(b.amt)  
FROM sales a  
JOIN sales b  
  ON b.date ≤ a.date  
GROUP BY a.date;
```



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Q10

"TOP 1" JOIN

Finding Max without Aggregates



Standard SQL

-- Row where no greater value exists

```
SELECT a.*  
FROM items a  
LEFT JOIN items b  
  ON b.price > a.price  
WHERE b.price IS NULL;
```



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Q11

NULL KEY JOIN

Joining when IDs can be NULL



Standard SQL

-- Standard '=' fails on NULLs

```
SELECT * FROM table_a a
JOIN table_b b
ON COALESCE(a.id, 0) = COALESCE(b.id, 0);
```



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Q12

LIKE JOIN

Matching partial strings



Standard SQL

-- Join log to domain rule

```
SELECT u.email, d.category
FROM users u
JOIN domains d
  ON u.email LIKE CONCAT('%', d.domain);
```



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Q13

DELETE DUPLICATES

Keeping only the first occurrence



MySQL / SQL Server

-- Self Join Delete

```
DELETE t1
FROM my_table t1
JOIN my_table t2
  ON t1.email = t2.email
WHERE t1.id > t2.id;
```



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Q14

MANY-TO-MANY

Using a Bridge/Junction Table



Standard SQL

-- Student -> Enrollment -> Class

```
SELECT s.name, c.class_name
FROM students s
JOIN enrollment e ON s.id = e.sid
JOIN classes c ON e.cid = c.id;
```



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RECURSIVE CTE

Finding all subordinates



Standard SQL

-- Join CTE back to Employee table

```
WITH RECURSIVE hier AS (  
  SELECT id FROM emp WHERE id = 1  
  UNION ALL  
  SELECT e.id FROM emp e  
  JOIN hier h ON e.mgr_id = h.id  
) SELECT * FROM hier;
```



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Q16

LATERAL JOIN

Join table to subquery PER ROW



PostgreSQL

-- Top 3 comments per user

```
SELECT u.name, c.text
FROM users u
CROSS JOIN LATERAL (
  SELECT text FROM comments
  WHERE user_id = u.id
  LIMIT 3
) c;
```



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Q17

CONDITIONAL JOIN

Join A or B based on Type



Standard SQL

-- Polymorphic Join Logic

```
SELECT e.event_name,  
        COALESCE(u.name, c.name)  
FROM events e  
LEFT JOIN users u  
    ON e.rel_id = u.id AND e.type = 'User'  
LEFT JOIN companies c  
    ON e.rel_id = c.id AND e.type = 'Co';
```



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Q18

HAVING VS WHERE

Filtering Aggregates



Standard SQL

-- Filter AFTER grouping

```
SELECT c.name, COUNT(o.id)
FROM customers c
JOIN orders o ON c.id = o.cid
GROUP BY c.name
HAVING COUNT(o.id) > 5;
```



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Q19

JOIN VS EXISTS

Checking presence efficiently



Standard SQL

-- Exists stops at first match

```
SELECT * FROM orders o
WHERE EXISTS (
  SELECT 1 FROM products p
  WHERE p.id = o.pid
  AND p.category = 'Tech'
);
```



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Q20

FAN-OUT TRAP

Accidentally exploding rows



Standard SQL

-- 1:Many joins duplicate left rows

SELECT c.name, **COUNT**(*)

FROM customers c

JOIN orders o **ON** c.id = o.cid

-- Result count is Orders, NOT Customers!

GROUP BY c.name;



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YOU MADE IT!



LEARNED A NEW PATTERN?

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